

Reproducing the `my_surface_iter` GP surface-bundle scaling loop

This note describes the **iterative GP surface scaling** workflow that produced the archived workspace `runs/my_surface_iter/`: alternating a full `run_gp` fit with spectroscopic **per-spec_bundle_id** linear rescaling against the GP latent surface on `X_fill` (default `mu_raw`).

Prerequisites

- **Python** 3.10+ recommended
- **Packages:** `numpy`, `scipy`, `matplotlib`, `george`
- Working directory: **repository root** (`gp/`), so imports and default `runs/` paths resolve correctly.

```
cd /path/to/gp
pip install numpy scipy matplotlib george
```

Inputs used for `my_surface_iter`

Role	File (paths relative to repo root)
Starting training bundle	<code>gp_work_scaled_nophot_m8767_m8217.npz</code> (or whatever you pass as <code>--input-bundle</code>)
Grid / flux normalization meta	<code>gp_scaled_bundle_meta.json</code> (<code>grid_norm_info</code> required)

The driver copies the input bundle to `runs/my_surface_iter/iter_00/bundle.npz`.

What the driver does (high level)

1. For each outer iteration `k = 0 ... K-1`:
 - Runs `run_gp.py` on `runs/my_surface_iter/iter_{k:02d}/bundle.npz` with tag `surf_iter_k{k:02d}` under `runs/` (or your `--runs-dir`).
 - Copies `predictions.npz` and `config.json` into the workspace iteration folder.
 - Interpolates `mu_raw` (or `mu`) from `X_fill` onto spectroscopic training rows, solves a **closed-form WLS** linear scale per `spec_bundle_id`, clips multipliers, and applies them via `bundle_scale_pipeline.apply_epoch_linear_multiplier` (photometry unchanged). Writes `iter_{k+1}/bundle.npz`.
2. For `k >= 1`, `run_gp` is warm-started from `runs/<gp-tag-prefix>_k{k-1:02d}/config.json` when present (`--warm-start-config-json`).
3. Appends one JSON line per iteration to `iteration_log.jsonl`; after the loop, writes `metrics/*.png`, `scaling/` CSVs and evolution plots for highlight bundles (default 3 and 5 if `--diag-bundles` omitted).

Command that matches the archived my_surface_iter layout

Tags under `runs/` are `surf_iter_k00 ...` (prefix `surf_iter`). Workspace directory `my_surface_iter`. Adjust paths if your clone lives elsewhere.

Important: Flags `--diag-bundles`, `--diag-full-overview-interval`, `--plot-results-each-iter` belong to `iterate_gp_surface_bundle_scale.py` and must appear **before** a lone `--`. Only `run_gp.py` options go **after** `--`.

```
cd /path/to/gp
```

```
python3 iterate_gp_surface_bundle_scale.py \  
  --input-bundle gp_work_scaled_nophot_m8767_m8217.npz \  
  --meta gp_scaled_bundle_meta.json \  
  --workspace runs/my_surface_iter \  
  --runs-dir runs \  
  --gp-tag-prefix surf_iter \  
  --max-iters 20 \  
  --run-gp-max-iter 60 \  
  --surface-mu-key mu_raw \  
  --diag-bundles 3,5 \  
  --diag-full-overview-interval 5 \  
  --plot-results-each-iter \  
  --converge-max-log-scale 5e-4 \  
  --converge-delta-chi2-spec 5e-4 \  
  --bundle-scale-clip 10 \  
  -- \  
  --additive-time --additive-wls \  
  --kernel-time matern52 --kernel-wls matern52 \  
  --mean linear --optimize \  
  --meta-json gp_scaled_bundle_meta.json
```

Lighter run (smoke test)

```
python3 iterate_gp_surface_bundle_scale.py \  
  -i gp_work_scaled_nophot_m8767_m8217.npz \  
  --meta gp_scaled_bundle_meta.json \  
  -w runs/smoke_surface_iter \  
  --gp-tag-prefix smoke_surf \  
  --max-iters 1 \  
  --run-gp-max-iter 10 \  
  --diag-bundles 3,5 \  
  -- \  
  --additive-time --additive-wls \  
  --kernel-time matern52 --kernel-wls matern52 \  
  --mean linear --optimize \  
  --meta-json gp_scaled_bundle_meta.json
```

Where outputs land

Location	Contents
runs/my_surface_iter/iter_XX/bundle.npz	Working bundle per outer iteration
runs/my_surface_iter/iter_XX/predictions.npz	Copy of GP predictions for that iteration
runs/my_surface_iter/iter_XX/config.json	Copy of <code>run_gp</code> config + metrics
runs/my_surface_iter/iteration_log.jsonl	One JSON object per iteration (scales, chi-squared, kernel dict, etc.)
runs/my_surface_iter/metrics/	Summary plots after the outer loop finishes (<code>chi2_vs_iter.png</code> , ...)
runs/my_surface_iter/scaling/	Per-epoch cumulative scale CSVs + PNGs for highlight bundles
runs/my_surface_iter/iter_XX/figs/overview	Targeted <code>plot_bands_gp_overview</code> when <code>--diag-bundles</code> is set
runs/my_surface_iter/iter_XX/figs/overview_full	Full overview when <code>--diag-full-overview-interval > 0</code>
runs/surf_iter_kXX/	Each full <code>run_gp</code> run (predictions, config, figs/ if <code>plot_results</code> was invoked)

Post-processing: scales, standard plots, overview

Use the **same** `--runs-dir` / `--output-dir` as `run_gp` if you did not use the repo default (see `plot_results.py` help). `GP_TAG` must match `run_gp -t`.

```
# Kernel / length-scale summary from a finished surf_iter tag
python3 gp_scales.py surf_iter_k19
```

```
# Standard diagnostics for one outer iteration's GP tag
python3 plot_results.py \
  --tag surf_iter_k19 \
  --output-dir "$(pwd)/runs" \
  --bundle runs/my_surface_iter/iter_19/bundle.npz \
  --meta gp_scaled_bundle_meta.json \
  --heatmap-raw
```

```
# Band + bundle overview (train posterior)
python3 plot_bands_gp_overview.py \
  --bundle runs/my_surface_iter/iter_19/bundle.npz \
  --meta gp_scaled_bundle_meta.json \
  --tag surf_iter_k19 \
  --runs-dir "$(pwd)/runs" \
  --expect-pipeline-bundle
```

```
# Optional: raw fill-grid mu panel
python3 plot_bands_gp_overview.py \
  --bundle runs/my_surface_iter/iter_19/bundle.npz \
  --meta gp_scaled_bundle_meta.json \
  --tag surf_iter_k19 \
  --runs-dir "$(pwd)/runs" \
```

```
--expect-pipeline-bundle \  
--posterior-kind grid_raw \  
--output-dir runs/my_surface_iter/iter_19/figs/overview_grid_raw
```

Notes on bundle 3 and fit quality

Residual structure in **spec_bundle_3_*** panels can reflect **intrinsic data / cross-arm scaling limits** as much as kernel hyperparameters. If chi-squared and bundle multipliers stabilize but one bundle still disagrees with the GP slice, treat that as a **data / pipeline** hypothesis until a narrower wavelength kernel or different preprocessing is justified.

Reproducibility archive

The zip **my_surface_iter_repro_bundle.zip** (produced next to this doc in the repo) contains:

- **runs/my_surface_iter/** — full workspace (all **iter_***, logs, metrics, scaling, figs)
- **gp_scaled_bundle_meta.json** — meta used with the scaled bundle
- This markdown and its PDF copy

Re-running the **exact** outer loop from scratch still requires the **rest of this repository** (**iterate_gp_surface_bundle_scale.py**, **run_gp.py**, **bundle_scale_pipeline.py**, **gp_utils.py**, ...) and the **original** scaled NPZ if you do not rely on **iter_00/bundle.npz** inside the workspace.

Optional: to archive raw **run_gp** directories as well (not required to inspect the saved workspace), add **runs/surf_iter_k*** to a separate zip (~tens of MB per 20 iterations).